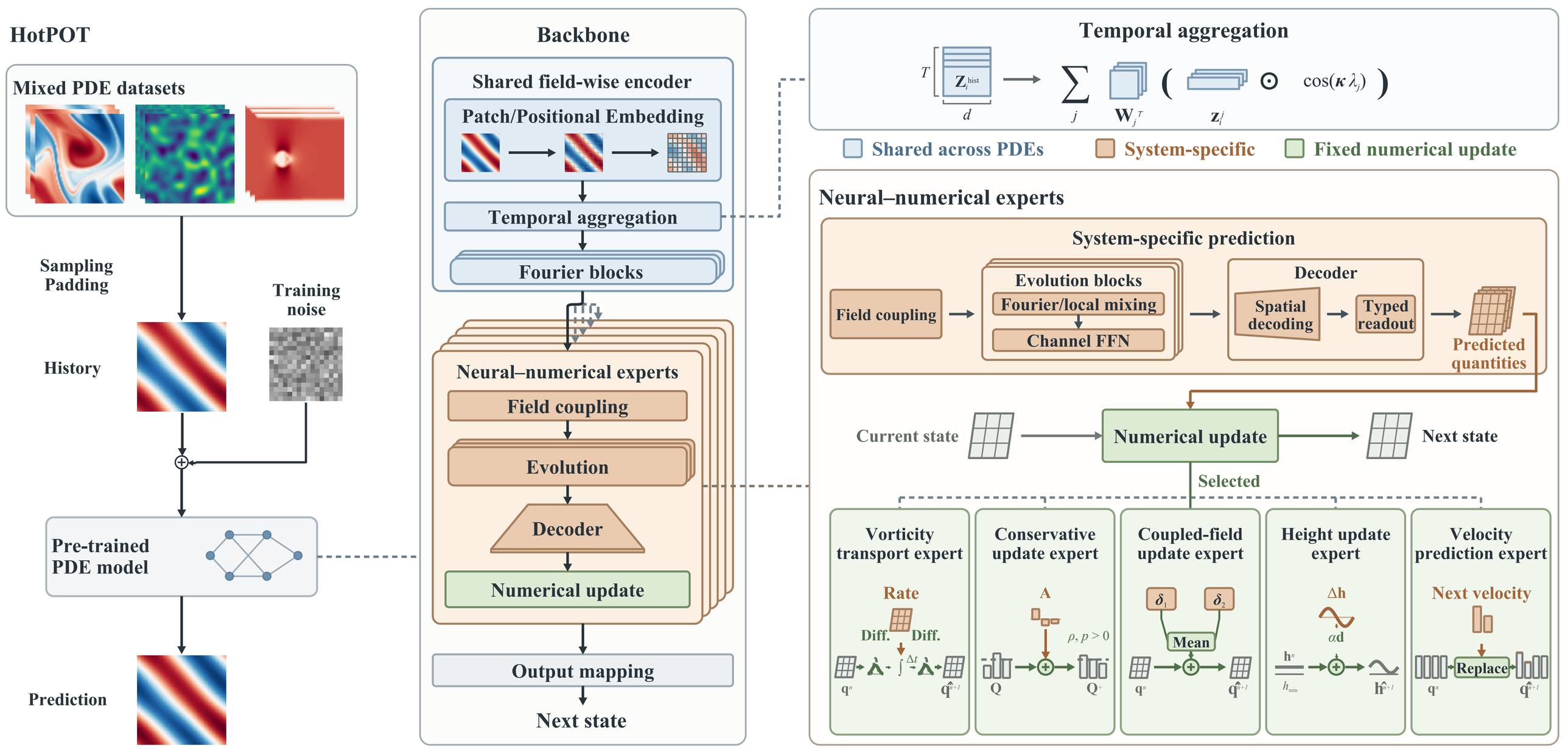

HotPOT
Mixed PDE datasets
Sampling
Padding
Training
noise
History
Pre-trained
PDE model
Prediction
Backbone
Shared field-wise encoder
Patch/Positional Embedding
Temporal aggregation
Fourier blocks
Neural–numerical experts
Field coupling
Evolution
Decoder
Numerical update
Output mapping
Next state
Temporal aggregation
Shared across PDEs
System-specific
Fixed numerical update
Neural–numerical experts
System-specific prediction
Field coupling
Evolution blocks
Fourier/local mixing
Channel FFN
Decoder
Spatial
decoding
Typed
readout
Predicted
quantities
Current state
Numerical update
Next state
Selected
Vorticity
transport expert
Rate
Diff.
Diff.
Conservative
update expert
A
ρ, p > 0
Coupled-field
update expert
Mean
Height update
expert
Δh
αd
Velocity
prediction expert
Next velocity
Replace